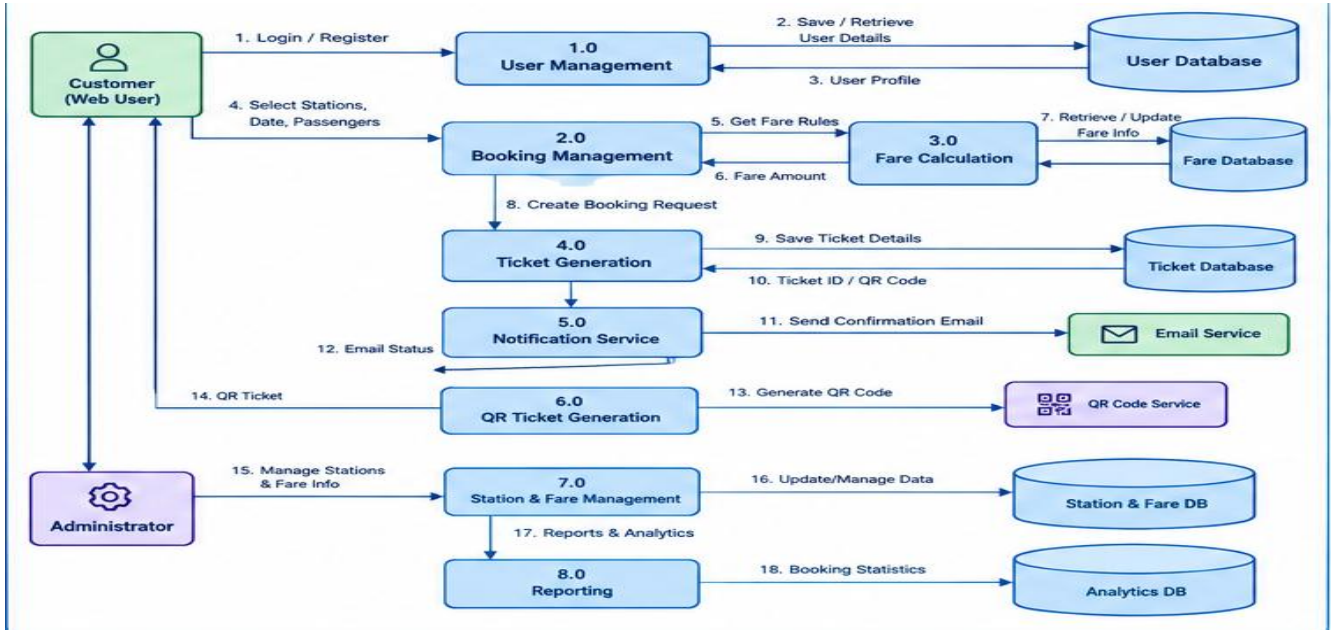
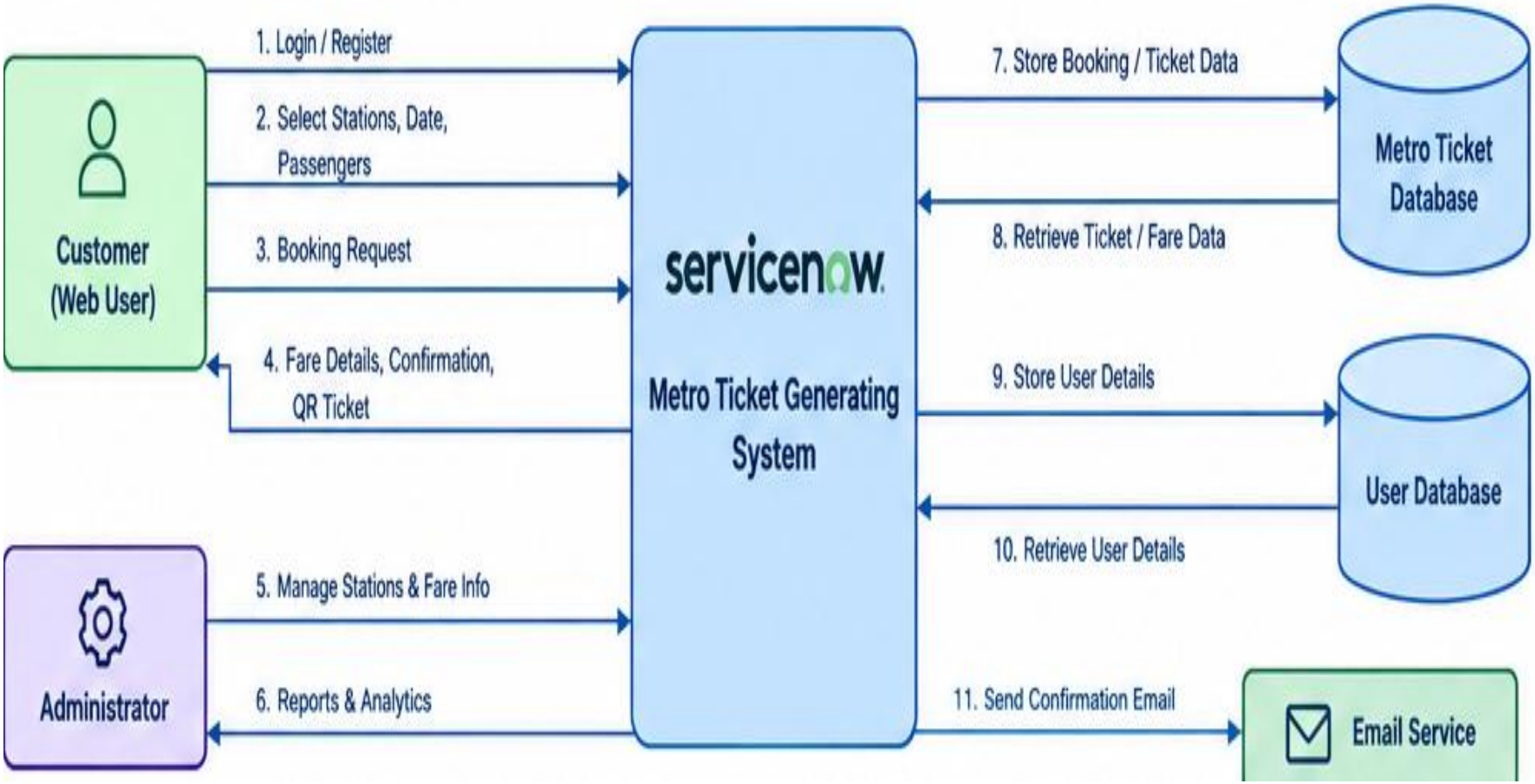
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	01-10-2026
Team ID	23KD1A4233
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:




1. Login / Register
2. Select Stations, Date, Passengers
3. Booking Request
4. Fare Details, Confirmation, QR Ticket
5. Manage Stations & Fare Info
6. Reports & Analytics
7. Store Booking / Ticket Data
8. Retrieve Ticket / Fare Data
9. Store User Details
10. Retrieve User Details
11. Send Confirmation Email
Customer (Web User)
Administrator
servicenow
Metro Ticket Generating System
Metro Ticket Database
User Database
Email Service

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Commuter (Portal User)	Route Selection & Origing	USN-1	As a commuter, I want to choose my origin and destination stations from an interactive dropdown so that I can define my travel path.System validates that selected origin and destination stations are active and distinct routes..	System validates that selected origin and destination stations are active and distinct routes.	High	Sprint-1
Commuter (Portal User)	Schedule & Passenger Info	USN-2	As a commuter, I want to specify my travel date, journey time slot, and passenger count so that I can reserve accurate pass slots.	Date selector restricts past dates, and passenger counter allows valid integer inputs (1–10).	High	Sprint-1
Commuter (Portal User))	Real-Time Fare Computation	USN-3	As a commuter, I want the system to calculate distance-based ticket costs automatically upon route selection so that I know the individual rate immediately.	Fare is rendered dynamically without requiring a manual page refresh or form submission.	High	Sprint-1

Commuter (Portal User)	Total Cost Summary	USN-4	As a commuter, I want to view an updated total price breakdown whenever passenger counts change so that I can confirm full fare before ordering.	Total fare instantly recalculates as $Total = BaseFare \times Passengers$ and updates on screen.	High	Sprint-1
Commuter (Portal User)	Pass Order Processing	USN-5	As a commuter, I want to submit my pass booking so that my order is registered into the system for ticket generation.	System generates a unique Request ID (REQ) and auto-creates an underlying Requested Item (RITM) record.	High	Sprint-1
Commuter (Portal User)	Pass Issuance Display	USN-6	As a commuter, I want to view a digital booking confirmation page containing my transaction reference number upon submission.	Confirmation screen displays booking reference ID, route summary, total price, and timestamp.	High	Sprint-1
Commuter (Portal User)	Automated Pass Dispatch	USN-7	As a commuter, I want to receive an automated email confirmation containing my pass summary so that I have a portable proof of booking.	System triggers a notification email with route details and a link to view the ticket.	Medium	Sprint-2
Commuter (Portal User)	Encrypted QR Pass Creation	USN-8	As a commuter, I want to receive a digital QR code linked to my pass so that I can scan it at metro entry gates.	Unique QR code image is generated, attached to the RITM record, and rendered on the user portal.	Medium	Sprint-2

System Admin	Station Network Control	USN-9	As an administrator, I want to create, modify, and deactivate metro stations in the master directory so that station data stays up to date.	Station master table permits full CRUD operations with mandatory name and station code	High	Sprint-1
System Admin	Tariff Schedule Management	USN-10	As an administrator, I want to update origin-destination fare matrices so that ticket pricing accurately reflects updated metro tariffs.	Admin can update rates per route pair, and updated values immediately apply to new bookings.	High	Sprint-1
System Admin	Analytics & Usage Insights	USN-11	As an administrator, I want access to executive dashboards tracking ticket volume, peak revenue routes, and daily sales trends.	Interactive reporting dashboard renders real-time visual metrics and downloadable ticket logs.	Medium	Sprint-2